

15.5 Floods

LEARNING INTENTION

By the end of this subtopic, you will be able to identify the causes of floods and understand where flooding is more likely to take place.

15.5.1 Causes of flooding

Even though Australia is the driest of all the world's inhabited continents, there are periods of very heavy rainfall and **flood**. Flood disasters in Australia damage property, kill livestock, and cause the loss of human life. Since 1788 more than 2000 people have died in floods, equalling the number of deaths from cyclones. In some cases, entire sections of a town have been washed away, as in 1852, when one-third of the town of Gundagai disappeared.

Floods are an unusual accumulation of water that overflows from rivers, lakes or the ocean onto land that is not normally covered by water. The flooding can be caused by a number of factors including storms, and low-pressure systems such as cyclones, typhoons and hurricanes. Heavy rainfall during storms can lead to a rapid rise in water level over a very short period of time. This is known as flash flooding. The water is unable to soak into the ground quickly, and so it runs down to the river and rushes towards the sea. Fast-moving floodwater (**FIGURE 1**) causes great risk to property and human life because there is less time to prepare.

FIGURE 1 Flash flood in Toowoomba, Queensland, 2011



FIGURE 2 Damage left in Toowoomba, 2011



flood an unusual accumulation of water onto land that is not normally covered by water

Riverine flooding occurs when rivers burst their banks and cover the surrounding land. It is a relatively slower process and may isolate communities for many weeks. There are other forms of flooding including those caused by storm surges and tsunamis that lead to sea water flooding.

15.5.2 Floods and floodplains

Floods are a natural occurrence, but they are a natural hazard to humans, who tend to build farms, towns and transport routes in areas such as floodplains. A **floodplain** is an area of relatively flat land that borders a river and is covered by water during a flood. Floodplains are formed when the water in a river slows down in flat areas. The river begins to meander and gradually deposits **alluvium**, which builds up the floodplain and other landforms such as deltas.

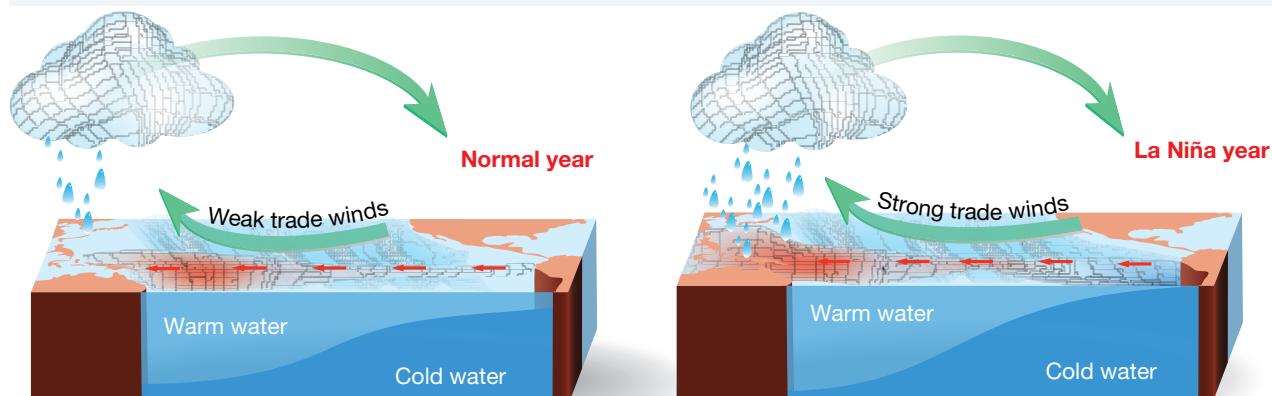
These fertile, flat areas are used for farming and settlement around the world. In Australia, many of our richest farmlands are on floodplains, and towns are often built on them because they are close to rivers. Such towns are subject to flooding. The possibility of flood is also increased when vegetation in **catchment areas** has been cleared or modified. Native vegetation can slow down run-off and reduce the chance of flooding.

15.5.3 La Niña and floods

A **La Niña** event in Australia is often associated with floods. Very cold waters dominate the eastern Pacific near South America, and the oceans off the coast of Australia become warmer than normal. As a result, large areas of low-pressure extend over much of Australia; warm, moist air moves in; and above-average rainfall occurs. There can also be torrential rain, widespread floods and tropical cyclones.

riverine relating to rivers
floodplain relatively flat land by a river
alluvium the loose material brought down by a river and deposited on its bed, or on the floodplain or delta
catchment area the area of land that contributes water to a river and its tributaries
La Niña a pattern in the climate that includes warmer ocean waters off Australia's east coast

FIGURE 3 A La Niña event leads to more rain in Australia.



During 2010–2011, many parts of Queensland, New South Wales and Victoria were flooded (**FIGURE 4**) during a La Niña event. Further severe flooding took place during March 2021. Historical flood levels were recorded across south-eastern Australia during this flooding event, which was influenced by a La Niña weather phase.

FIGURE 4 Rockhampton, Queensland (a) before and (b) during the 2011 flood



CASE STUDY

NSW floods March 2021

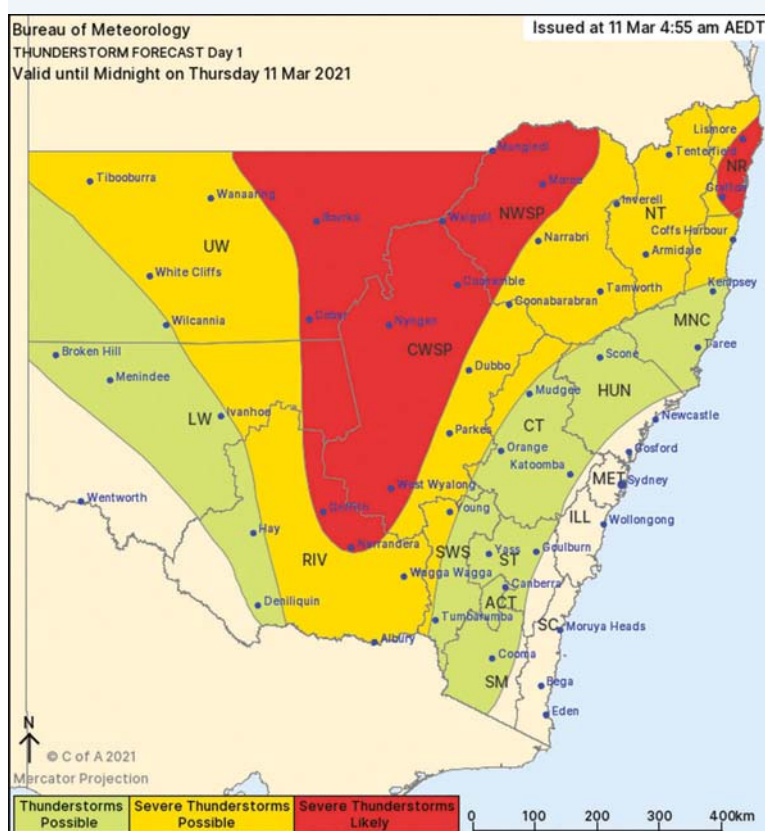
Many parts of Australia experienced intense weather patterns during February and March 2021, which resulted in historical flooding and rainfall records. In late February, severe storms in south-east Queensland generated large hailstones, flash flooding and widespread power outages. A series of storms continued into early March across northern NSW.

On 11 March, widespread thunderstorm warnings were issued across central and northern NSW (**FIGURE 5**). Slow-moving weather systems generated heavy and sustained rainfall across NSW over the following week. River catchments across the state received heavy rainfall totals leading to flash flooding and historic rainfall totals. At Mount Seaview west of Port Macquarie, 815 millimetres of rain fell between 19 and 23 March, and inland at Moree the Mehi River peaked at 10.38 metres on 25 March. (View the boundaries of NSW river catchments using your **NSW Catchments** weblink in your online Resources.)

As heavy rainfall continued to fall within the Hawkesbury-Nepean River catchment, western and north-western suburbs of Sydney faced widespread flooding and evacuations.

The Warragamba Dam, upstream of the suburbs near Penrith, Richmond and Windsor, reached full capacity on 21 March. As further rain fell within the catchment, the water of Warragamba Dam started to spill over the dam wall, contributing to higher river levels downstream. Major road networks were cut off by the flood waters and property was destroyed. The 'bathtub effect' of flooding within the Hawkesbury-Nepean catchment is demonstrated in **FIGURE 10**.

FIGURE 5 An alert issued on 11 March 2021 for widespread thunderstorm activity.



Source: Reproduced by permission of Bureau of Meteorology, ©2021 Commonwealth of Australia.

FIGURE 6 Flood waters cut off access for motorists and residents living in Windsor.



FIGURE 7 Properties along the Hawkesbury-Nepean River were damaged by rising floodwaters.



FIGURE 8 The flood indicator at Windsor indicating 13 metre flood levels

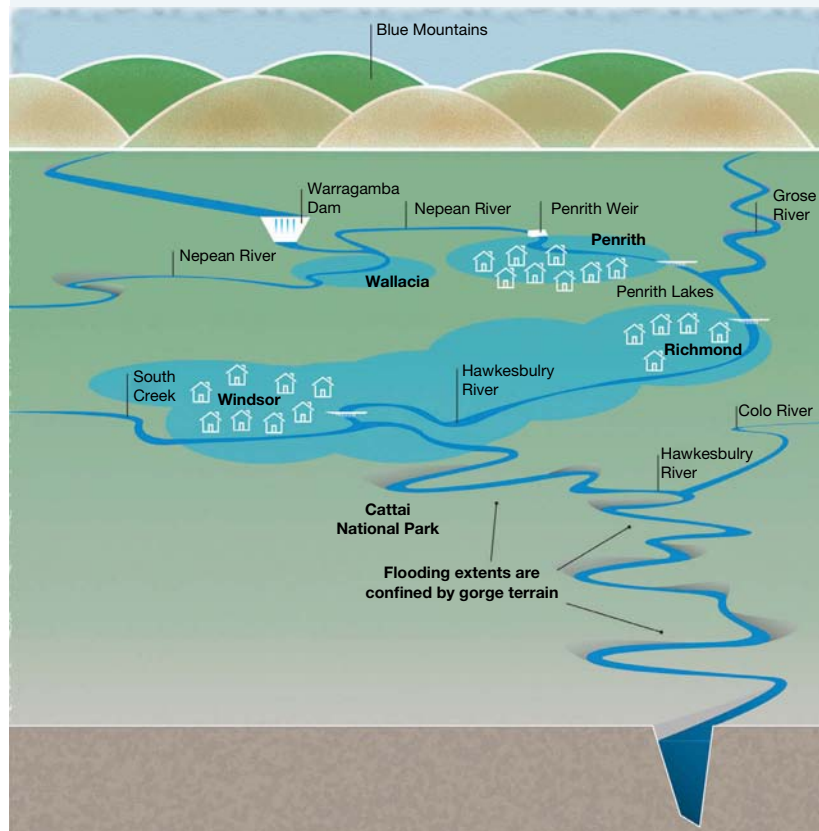


FIGURE 9 Windsor, a semi rural suburb of Sydney, (a) before and (b) during the 2021 flood



Source: <https://earthobservatory.nasa.gov/images/148093/historic-floods-in-new-south-wales>

FIGURE 10 The ‘bathtub’ effect of flooding within the Hawkesbury-Nepean River catchment



Source: © NSW Government

Resources

-  **eWorkbook** Floods (ewbk-9494)
-  **Video eLesson** Floods — Key concepts (eles-5126)
-  **Interactivities** Floods (int-8551)
2011 floods in Rockhampton (int-5291)

15.5 ACTIVITIES

1. Make a sketch of a local river catchment near you. Using words such as *floodplain*, *tributary* and *watershed*, annotate your diagram to explain how rivers make their way to the sea.
2. Research and write a news-style report on a recent flood in Australia. (You could use the case study of the NSW 2021 floods as a starting point for the kinds of information you might collect.) Using the **El Niño–Southern Oscillation (ENSO)**, **El Niño — What is it?** and **The impacts of La Niña in Australia** weblinks in your Resources panel, include information about the impact of El Niño and La Niña on the frequency and severity of Australian flood events.
3. Create an overlay map to show the distribution of water in the aerial photos of Rockhampton or Windsor before and after the 2011 or 2021 floods. Estimate the percentage of the area covered in both maps. Describe the changes to the environment.
4. Collect a media source that describes the impact of the high rainfall received in March 2021 on one location within NSW that experienced flooding.
 - a. Identify the catchment that the area is located within.
 - b. Outline the impact of the high rainfall on this location.
 - c. Suggest two reasons why this location flooded.

15.5 EXERCISE

Learning pathways

■ LEVEL 1

1, 2, 8, 12, 15

■ LEVEL 2

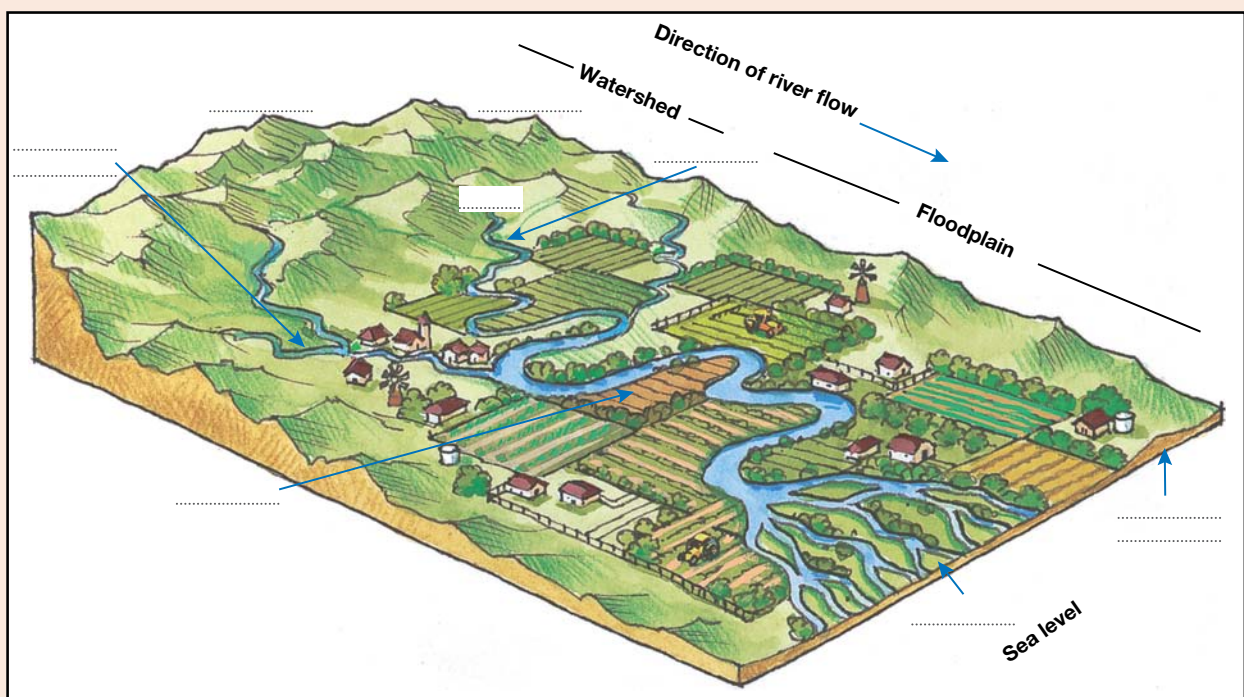
4, 6, 7, 9, 13

■ LEVEL 3

3, 5, 10, 11, 14

Check your understanding

1. What is a flood?
2. What is a floodplain?
3. Why is alluvium important to agriculture?
4. Construct a simple flow diagram to show the process of a La Niña event.
5. What does the term *La Niña* mean?
6. Describe how floodplains form.
7. Label the missing elements on the following diagram.



Apply your understanding

8. Create an infographic warning potential homeowners about the consequences of buying a house in floodplain regions.
9. Why do floods occur on floodplains and in deltas?
10. How might the effects of floods in urban spaces differ from effects in rural spaces?
11. Explain how native vegetation might help to reduce flooding.
12. Explain one benefit and one possible danger of farming on a floodplain.

Challenge your understanding

13. Should people continue to build on floodplains? Why or why not?
14. Climate change is warming overall ocean temperatures. Predict how this will affect how often Australia experiences La Niña conditions.
15. Suggest one strategy that farmers might use to protect their riverside properties from flooding.

To answer questions online and to receive **immediate feedback** and **sample responses** for every question, go to your learnON title at www.jacplus.com.au.